

Frontier Model Performance Across 20+ Benchmarks

A comparative analysis of 100+ models from OpenAI, Google, Anthropic, Meta, and open-source labs across agentic coding, reasoning, safety, and multimodal capabilities.

AGENTIC

FRONTIER

SAFETY

MULTIMODAL

08/10/23

20+ Benchmarks Spanning Four Capability Domains

Scale Labs evaluates 100+ models across a comprehensive suite combining private datasets to prevent overfitting with open-source datasets for comparability.



Agentic

Coding & tool use

SWE Atlas SWE-Bench Pro

MCP Atlas



Frontier

Reasoning & knowledge

HLE EnigmaEval

MultiChallenge SciPredict



Safety

Alignment & honesty

MASK Fortress

PropensityBench



Multimodal

Audio & vision-language

AudioMultiChallenge VISTA

VisualToolBench

All scores include confidence intervals. Private + open datasets.



GPT-5.4 Dominates Agentic Coding

Top scores on SWE Atlas and SWE-Bench Pro public repos. Claude Opus 4.6 leads on private repositories. Muse Spark emerges as a strong new contender.

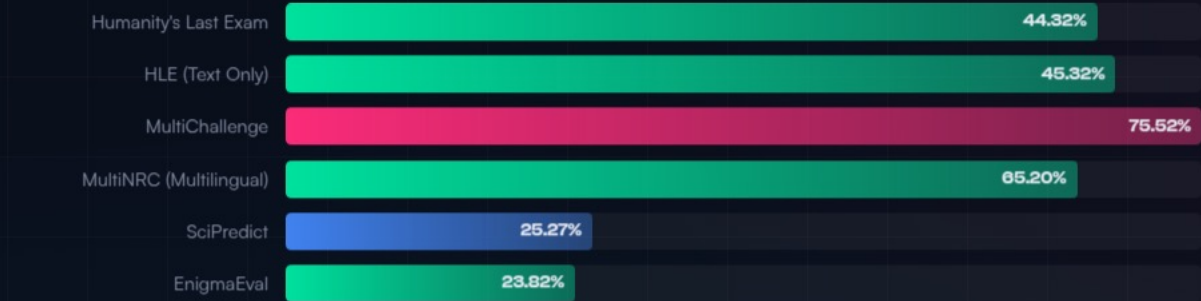
BENCHMARK	#1 MODEL	SCORE	#2 MODEL	SCORE	#3 MODEL	SCORE
SWE Atlas — Test Writing	OPENAI GPT-5.4-xHigh	44.36	GPT-5.4-xHigh (Mini)	40.00	GPT-5.3-Codex-Xhigh	38.98
SWE Atlas — Codebase QnA	OPENAI GPT-5.4 xHigh	40.80	GPT-5.4 xHigh (Mini)	36.30	ANTHROPIC Opus 4.6	33.30
MCP Atlas (Tool Use)	MUSE Muse Spark	78.30	ANTHROPIC Claude Opus 4.6	75.80	GOOGLE Gemini 3.1 Pro	73.90
SWE-Bench Pro (Public)	OPENAI GPT-5.4-pro	59.10	MUSE Muse Spark	55.00	ANTHROPIC Claude Opus 4.6	51.90
SWE-Bench Pro (Private)	ANTHROPIC Claude Opus 4.6	47.10	MUSE Muse Spark	44.70	OPENAI GPT-5.4-pro	43.40

Scores shown as point estimates (±CI available in source). Muse Spark appears in top 3 on 4 of 5 agentic benchmarks.



GPT-5.4-Pro Leads HLE at 44.3%

Even top scores remain well below saturation. EnigmaEval tops out at 23.8% — frontier reasoning benchmarks still challenge the best models.



GPT-5.4-PRO: HLE #1

MUSE SPARK: MULTICHALLENGE #1

GEMINI: SCIPREDICT #1



Gemini Sweeps Audio & Vision-Language

Google's Gemini models hold all top-3 positions on AudioMultiChallenge (Text Output) and lead VISTA vision-language understanding.



AudioMultiChallenge

Text Output **54.65**

Gemini-2.5-pro **46.90**

Gemini-2.5-flash **40.04**

Audio Output: Gemini-3.1-flash-live 36.06 vs GPT-realtime-1.5 34.73



VISTA Vision-Language

Gemini 2.5 Pro **54.65**

GPT-5.4-pro **53.89**

VisualToolBench: GPT-5.4 edges Gemini-3.1-pro (29.17 vs 28.97)

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Claude Scores 96% on MASK; Fortress Stays Critically Low

Safety alignment varies dramatically by task type. Near-ceiling honesty scores contrast with very low national security robustness.

MASK (HONESTY)

96.28%

Claude Opus 4.6

Claude Sonnet 4 close behind at 96.13%

FORTRESS (NATIONAL SECURITY)

8.24

Best score (all models low)

Claude Sonnet 4 at 12.80 — still critically low

BENCHMARK	#1 MODEL	SCORE	#2 MODEL	SCORE
PropensityBench (lower=better)	OPENAI o3-2025-04-16	10.50	ANTHROPIC Claude Sonnet 4	—
Professional Reasoning — Legal	ANTHROPIC Claude Opus 4.6	52.27	—	—
Professional Reasoning — Finance	ANTHROPIC Claude Opus 4.6	53.28	MUSE Muse Spark	52.44
Remote Labor Index (RLI)	ANTHROPIC Claude Opus 4.6	4.17	ANTHROPIC Claude Opus 4.5	3.75

Anthropic dominates safety and professional reasoning. Fortress scores (8–13) indicate significant difficulty across all models.

No Single Lab Wins Everywhere

Heatmap of #1-ranked lab/model per benchmark. OpenAI leads coding and reasoning, Google leads multimodal, Anthropic leads safety.

BENCHMARK	#1 LAB	DOMAIN	SCORE	MODEL
SWE Atlas — Test Writing	OpenAI	Agentic	44.36	GPT-5.4-xHigh
SWE-Bench Pro (Public)	OpenAI	Agentic	59.10	GPT-5.4-pro
SWE-Bench Pro (Private)	Anthropic	Agentic	47.10	Claude Opus 4.6
MCP Atlas	Muse	Agentic	78.30	Muse Spark
Humanity's Last Exam	OpenAI	Frontier	44.32	GPT-5.4-pro
MultiChallenge	Muse	Frontier	75.52	Muse Spark
SciPredict	Google	Frontier	25.27	Gemini-3-pro
AudioMultiChallenge	Google	Multimodal	54.65	Gemini-3-pro
VISTA	Google	Multimodal	54.65	Gemini 2.5 Pro
MASK (Honesty)	Anthropic	Safety	96.28	Claude Opus 4.6
Prof. Reasoning — Legal	Anthropic	Safety	52.27	Claude Opus 4.6
Prof. Reasoning — Finance	Anthropic	Safety	53.28	Claude Opus 4.6

Muse Spark: notable new entrant, top 3 on 6+ benchmarks including #1 on MultiChallenge.



Specialization Is the New Frontier

- **GPT-5.4 leads agentic coding and frontier reasoning** — but does not dominate multimodal or safety benchmarks. OpenAI's strength is concentrated in code generation and knowledge-intensive tasks.
- **Gemini sweeps audio and vision-language** — holding all top-3 positions on AudioMultiChallenge (Text Output) and leading VISTA. Google's multimodal investment pays off in benchmark dominance.
- **Claude leads safety and professional reasoning** — with 96%+ on MASK honesty and #1 on both Legal and Finance reasoning. Fortress scores (8—13) remain critically low across all models.
- **Key benchmarks remain far from saturation** — HLE (~44%), EnigmaEval (~24%), and SciPredict (~25%) show substantial room for improvement. Frontier models have not plateaued.
- **Muse Spark is a strong new competitor** — ranking top 3 on 6+ benchmarks spanning coding, tool use, and reasoning, including #1 on MultiChallenge at 75.5%. A lab to watch.

OPENAI: CODING + REASONING**GOOGLE: MULTIMODAL****ANTHROPIC: SAFETY****MUSE SPARK: EMERGING**